

Lecture 07 : IP addressing and routing (Part II)

Introduction

In the vast interconnected landscape of the internet, every device needs a unique identifier to communicate effectively. Just as a physical address guides mail to your home, an Internet Protocol (IP) address directs data packets to their correct destination. This lecture delves into the fundamental concepts of IP addressing, building upon previous discussions of packet fragmentation and reassembly. We will explore the structure of IP addresses, their hierarchical nature, and the classification system that organizes them, known as IP address classes. Understanding these concepts is crucial for anyone seeking to grasp the foundational mechanisms of network communication and how data finds its way across the global network of networks. Without a unique and structured addressing system, the internet as we know it would simply cease to function, leading to chaos and an inability to deliver information reliably. This session will illuminate how this critical uniqueness and organization are achieved, laying the groundwork for more advanced networking topics.

Core Concepts

Term	Definition	Significance
IP Address	A unique 32-bit numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.	Ensures that data packets can be uniquely routed to the correct destination device across the internet, preventing confusion and misdelivery.
Dotted Decimal Notation	A human-readable format for expressing a 32-bit IP address, where the 32 bits are divided into four 8-bit segments (octets), each converted to its decimal equivalent and separated by dots.	Simplifies the representation and memorization of complex binary IP addresses, making them more practical for human use and configuration.
	The initial portion of an IP address that identifies the	Routers primarily use this part of the IP address to forward

Term	Definition	Significance
Network Number	specific network to which a host belongs.	packets to the correct destination network, enabling efficient inter-network communication.
Host Number	The latter portion of an IP address that uniquely identifies a specific device (host) within a particular network.	Allows local network administrators to assign unique identifiers to devices within their network, ensuring precise delivery of packets once they reach the target network.
IP Address Classes	A system of categorizing IP addresses (Class A, B, C, D, E) based on their leading bits, which determines the default division between network and host parts.	Provides a structured approach to allocating IP addresses, optimizing their use for networks of varying sizes and for different communication purposes (e.g., unicast, multicast).
Unicast Address	An IP address assigned to a single network interface, used for one-to-one communication.	Enables direct communication between two specific devices, forming the basis of most internet traffic.
Multicast Address	An IP address (specifically Class D) used to send a single packet to a select group of devices simultaneously.	Facilitates efficient group communication, such as video conferencing or streaming, by delivering data to multiple recipients without sending individual copies.
Private IP Address	A range of IP addresses reserved for use within private networks, not routable on the public internet.	Allows organizations to use a large number of IP addresses internally without consuming public IP address space, enhancing security and simplifying network management.

Term	Definition	Significance
Loopback Address	A special IP address (127.x.x.x) that refers to the local machine itself, used for testing network applications.	Provides a mechanism for a device to send network traffic to itself, useful for diagnostics and developing network services without involving external network hardware.
Broadcast Address	An IP address used to send a packet to all devices within a specific network (limited broadcast) or to all devices on the internet (directed broadcast, though less common).	Enables a device to communicate with all other devices on its local network segment, useful for discovery services or sending general announcements.

Detailed Analysis

1. The Fundamentals of IP Addressing

The Internet Protocol (IP) is a cornerstone of the TCP/IP family, operating at the network layer to ensure that every device connected to the internet possesses a unique identifier. This uniqueness is paramount; without it, routers would be unable to distinguish between multiple devices sharing the same address, leading to communication failures. An IP address is defined as a 32-bit quantity, allowing for approximately 2^{32} , or about 4 billion, distinct addresses.

To make these 32-bit binary numbers (e.g., 01011011...) more manageable for humans, the dotted decimal notation was introduced. This system divides the 32 bits into four 8-bit segments, known as octets. Each octet is then converted into its decimal equivalent, and these four decimal numbers are separated by dots. For example, a 32-bit binary address might be represented as 66.134.48.126 in dotted decimal notation, which is significantly easier to read, write, and remember.

2. Hierarchical Addressing and Routing

Every 32-bit IP address is fundamentally divided into two parts: a network number and a host number. This hierarchical structure mirrors how physical addresses are organized (e.g., country, city, street, house number). The first part, the network number, identifies the specific network to which a device belongs. Since the internet is a "network of networks," each network must have a globally unique network

number, managed by a central authority. The second part, the host number, uniquely identifies a particular device within that specific network. Local network administrators can manage host numbering within their assigned network number space.

This division is critical for routing. When a router needs to forward a packet, its primary task is to send the packet to the correct destination network. Therefore, routers only examine the network number part of the IP address. Once the packet arrives at the correct network, the router within that network uses the host number to deliver the packet to the precise destination device. This two-tiered approach optimizes routing efficiency by reducing the amount of information routers need to process at each hop.

3. IP Address Classes (Classful Addressing)

To manage the allocation and interpretation of IP addresses, a system of "classful addressing" was historically established, defining five IP address classes: A, B, C, D, and E. Classes A, B, and C are primarily used for assigning unicast addresses to individual computers. Class D is reserved for multicast communication, where a single packet is sent to a group of devices simultaneously. Class E is reserved for experimental or special purposes. The class of an IP address can be determined simply by looking at its first few bits.

4. Class A Addresses

Class A addresses are designed for very large networks. They are identified by a leading 0 bit.

- * **Structure:** The first 8 bits define the network, and the remaining 24 bits define the host.

- * **Network Bits:** 7 bits (after the leading 0). This allows for $2^7 = 128$ possible networks. However, the all-zero network (0.x.x.x) is reserved, leaving 127 usable Class A networks.

- * **Host Bits:** 24 bits. This allows for 2^{24} combinations. After reserving two combinations (all 0s for the network address and all 1s for the broadcast address), each Class A network can support $2^{24} - 2$, or approximately 16 million, hosts.

- * **Address Range:** 0.0.0.0 to 127.255.255.255.

5. Class B Addresses

Class B addresses are suitable for medium to large-sized networks. They are identified by the leading bits 10.

- * **Structure:** The first 16 bits define the network, and the remaining 16 bits define the host.

- * **Network Bits:** 14 bits (after the leading 10). This allows for $2^{14} = 16,384$ possible networks.
- * **Host Bits:** 16 bits. After reserving two combinations, each Class B network can support $2^{16} - 2$, or approximately 65,534, hosts.
- * **Address Range:** 128.0.0.0 to 191.255.255.255 .

6. Class C Addresses

Class C addresses are intended for smaller networks. They are identified by the leading bits 110 .

- * **Structure:** The first 24 bits define the network, and the remaining 8 bits define the host.
- * **Network Bits:** 21 bits (after the leading 110). This allows for 2^{21} , or approximately 2 million, possible networks.
- * **Host Bits:** 8 bits. After reserving two combinations, each Class C network can support $2^8 - 2$, or 254, hosts.
- * **Address Range:** 192.0.0.0 to 223.255.255.255 .

7. Class D and E Addresses

- **Class D:** Identified by leading bits 1110 . These are multicast addresses, used when a packet needs to be sent to a group of devices simultaneously rather than a single host. The address range is 224.0.0.0 to 239.255.255.255 .
- **Class E:** Identified by leading bits 1111 . These addresses are reserved for future use or experimental purposes. The address range is 240.0.0.0 to 255.255.255.255 .

In terms of address distribution, Class A networks, though few in number, occupy the largest portion of the total IP address space due to their massive host capacity. Class B occupies the next largest, followed by Class C.

8. Special Purpose IP Addresses

Beyond the standard unicast, multicast, and reserved addresses, several IP address ranges are designated for special purposes:

- * **Private IP Addresses:** These addresses are used within private networks and are not routable on the public internet. They allow organizations to create internal networks without needing globally unique IP addresses for every device.
- * **Class A Private Range:** 10.0.0.0 to 10.255.255.255
- * **Class B Private Range:** 172.16.0.0 to 172.31.255.255 (16 distinct Class B networks)

* Class C Private Range: 192.168.0.0 to 192.168.255.255

When devices with private IP addresses need to communicate with the public internet, a Network Address Translator (NAT) or proxy server is typically used to translate their private addresses to a public one.

* **Loopback/Local Address:** Any address starting with 127 (e.g., 127.0.0.1) is a loopback address. This address refers to the local machine itself and is used for testing network applications or services without sending traffic out to the physical network. Packets sent to a loopback address never leave the local computer.

* **Default Network Address:** The address 0.0.0.0 represents the default network. It can be used to refer to the current network or as a placeholder for a default route.

* **Limited Broadcast Address:** An address consisting of all 1s in the host part (e.g., 255.255.255.255 for a local network broadcast) is used for limited broadcast. This sends a packet to all devices within the sender's current network segment.

9. Special Host Part Conventions

For Class A, B, and C networks, two specific host number combinations have special meanings:

* **All 0s in the Host Part:** This combination identifies the network itself. For example, 118.0.0.0 refers to network 118. This is particularly useful for routers to maintain their routing tables.

* **All 1s in the Host Part:** This combination signifies the broadcast address for that specific network. For example, 118.255.255.255 would broadcast a packet to all hosts within network 118.

These conventions allow IP routers to make informed decisions about where to forward packets, whether to a specific host, an entire network, or all hosts within a local segment.

Key Takeaways

- [] IP addresses are 32-bit unique identifiers for devices on the internet, enabling communication.
- [] Dotted decimal notation simplifies the representation of 32-bit binary IP addresses into four decimal octets.
- [] IP addresses are hierarchical, consisting of a network number (for routing between networks) and a host number (for identifying devices within a network).
- [] Routers primarily use the network number to forward packets to the correct destination network.

- [] IP address classes (A, B, C, D, E) categorize addresses based on their leading bits, determining the default network/host split.
- [] Class A is for very large networks, Class B for medium-to-large, and Class C for smaller networks.
- [] Class D addresses are used for multicast communication, while Class E is reserved.
- [] Private IP address ranges (e.g., 10.x.x.x, 192.168.x.x) are used for internal networks and are not routable on the public internet.
- [] The loopback address (127.x.x.x) refers to the local machine and is used for self-testing.
- [] All 0s in the host part denote the network address, and all 1s in the host part denote the broadcast address for that network.

Conclusion

This lecture has provided a comprehensive overview of IP addressing, from its fundamental role in ensuring unique device identification to the intricate system of IP address classes. We've seen how the 32-bit structure, combined with the human-friendly dotted decimal notation, forms the backbone of internet communication. The hierarchical division into network and host parts is crucial for efficient routing, allowing routers to quickly direct traffic to the correct network before a local administrator guides it to the specific host. Understanding the characteristics and ranges of Class A, B, and C addresses, as well as the special purposes of Class D (multicast) and Class E (reserved), is essential for comprehending how IP addresses are allocated and utilized. Furthermore, the discussion of private, loopback, and broadcast addresses highlights the flexibility and specialized functions built into the IP addressing scheme. This foundational knowledge of IP addressing and its classful model is indispensable for anyone working with computer networks, setting the stage for future explorations into higher-layer protocols like TCP and UDP, which build upon the IP layer to provide diverse functionalities and features.